MD-DGFLS: MULTI-DOMAIN DEEPFAKE FORENSIC VERIFICATION CERTIFICATE

IEEE TPAMI / CVPR Technical Forensics Specification v2.2-PROD | Cryptographic Audit Dossier

File Name:	51415b10.jpeg	**Target Domain:**	Face
Audit Timestamp:	2026-09-25 10:25:26 UTC	**SHA-256 Hash:**	34b3f53565e42004d05a37e8...
Forensic Verdict:	**VERIFIED AUTHENTIC MEDIA**	**Confidence:**	**61.7%**
Tamper Type:	Authentic Media Stream	**Bounding Box:**	None

Three-Stage Visual Forensics Pipeline

Stage 1: Source Image

Stage 2: Tamper Overlay

Stage 3: XAI Heatmap

Mathematical Innovations & Multi-Expert Telemetry

CDFAM (Cross-Domain Alignment)	Contrastive Score: 0.4876 (Margin: m=0.3)
ARCC (Entropy Calibration)	Dynamic Calibration Temperature: T = 1.113
TGHFL (Transformer-GAN Cross-Attention)	Cross-Attention Energy: 0.0005
MEFCN (Multi-Expert Consensus)	Face: 73.7% | Medical: 8.4% | Satellite: 5.9% | Regular: 6.0% | Natural: 6.0%
HLRN (Localization Refinement)	Coarse-to-Fine Refinement: 1922x2560 Native
Frequency Stream (ELA & Laplacian)	ELA Discrepancy: 0.841 | Sensor Noise Var: 1409.2

This certificate is cryptographically anchored and generated by the MD-DGFLS Autonomous Forensics Engine. Verified under IEEE TPAMI standards.